ECN 713 Spring 2026
Natalia Kovrijnykh

**Problem Set 2: Due on Tuesday, February 3, @noon**
**Asset Pricing**
(100 points total)

1. (30 points) Consider the following asset-pricing model in an endowment economy. There is measure one of identical households. Households are infinitely lived and maximize the expected discounted utility
$$E_0 \sum_{t=0}^{\infty} \beta^t u(c_t),$$
where $c_t$ is consumption at time $t$, $\beta \in (0,1)$, and the utility function $u(\cdot)$ is strictly increasing, differentiable and strictly concave.

   There is a single divisible asset ('tree') in fixed supply normalized to one. Dividends, $z$, take strictly positive values in the set $Z = \{z_1, z_2, ...z_N\}$, and follow a Markov process with transition function $P(z, z') = \Pr(z_{t+1} = z' | z_t = z)$.

   There is also a government in this economy that taxes dividends at the tax rate $\tau \in [0,1)$ that takes values in the set $T = \{\tau_1, \tau_2, ...\tau_k\}$, and follows a Markov process with transition function $Q(\tau, \tau') = \Pr(\tau_{t+1} = \tau' | \tau_t = \tau)$. Tax collections are returned to households through a lump-sum rebate.

   Each household is initially endowed with one unit of the asset. The asset is traded in each period, after the realization of dividends and taxes.

   (a) (7 points) Write the problem of a household in the recursive language. Obtain the first-order condition and the Envelope condition that characterize its solution. Write down the stochastic Euler equation in equilibrium.

   (b) (13 points) Consider now the following special case. Suppose that the utility function is logarithmic, $u(c) = \ln(c)$. Dividends are constant over time and equal to $\bar{z} > 0$. There are two states for taxation, $\tau_L = 0$ and $\tau_H = \bar{\tau}$, $\bar{\tau} \in (0,1)$. The probabilities of the Markov process for taxation are such that $Q(\tau_L, \tau_L) = Q(\tau_H, \tau_H) = \theta \in (0,1)$. Obtain the functional equation that pins down the equilibrium asset prices in each state. Provide intuitive interpretation.

   (c) (10 points) Economist $A$ claims that the asset price must always be higher in the state with the zero tax than in the state with the positive tax, while economist $B$ argues that the comparison depends on the persistence $\theta$. Who is right? Provide a formal argument and explain your answer intuitively.

2. (30 points) Consider the following version of the Lucas tree model. There is a single asset whose dividend is deterministic and grows at a constant (gross) rate $\gamma > 1$. The asset is in fixed supply normalized to one perfectly divisible share.

There is a continuum of measure one of identical agents with preferences given by

$$\sum_{t=0}^{\infty} \beta^t U(c_t),$$

where $c_t$ is consumption at time $t$, the function $U$ is strictly increasing, strictly concave, $C^2$, and $\beta \in (0,1)$. Agents are initially endowed with one unit of the asset.

(a) (6 points) Write down the problem of the representative agent in a recursive form. Derive the stochastic Euler equation evaluated in equilibrium.

(b) (9 points) Suppose that the utility function is CRRA,

$$U(c) = \begin{cases} \frac{c^{1-\sigma}}{1-\sigma}, & \text{if } \sigma > 0,\ \sigma \neq 1, \\ \ln c, & \text{if } \sigma = 1. \end{cases}$$

Derive the equilibrium asset price in closed form. How does the asset price change over time?

(c) (8 points) Economist $A$ claims that in an economy with higher growth, the equilibrium asset price is higher. Is she right? Provide a formal argument and explain your answer intuitively.

(d) (7 points) Derive the expected return of the asset in closed form. How does the return compare to $1/\beta$? How does it change with $\gamma$? Explain your answers intuitively.

3. (35 points). Consider the following version of the Lucas tree model with a risk of war. There is a single asset. In the absence of war, the dividend is constant over time, equal to $z$. War can start with probability $\pi \in [0,1)$ in each period, but can happen only once. The result of war is a permanent loss in output—dividends are reduced to $sz$ in all future periods, where $s \in (0,1)$. That is, dividends are $z$ until the war happens, and $sz$ in all periods after it happens.

There is a representative agent endowed with one share of the asset. The agent's preferences over consumption are given by

$$E_0 \sum_{t=0}^{\infty} \beta^t U(c_t),$$

where $U$ is strictly increasing and strictly concave, $c_t$ is consumption at time $t$, and $\beta \in (0,1)$.

(a) (6 points) Write down the recursive problem of the agent *after* the war happened. Derive the asset's after-war equilibrium price and expected return. Interpret your results intuitively.

(b) (9 points) Write down the recursive problem of the agent *before* the war started, and derive the corresponding stochastic Euler equation.

(c) (12 points) Suppose that $U(c) = \ln(c)$. What is the asset's before-war equilibrium price and expected return? How do they compare to those in an alternative economy without the risk of war? Explain your results intuitively.

(d) (8 points) How would the comparison of prices in part (c) change if

$$U(c) = \frac{c^{1-\sigma} - 1}{1 - \sigma},$$

where $\sigma \geq 0$ and $\sigma \neq 1$? Explain your answer intuitively.